Multimedia Signal processing

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February, 27 2015

1. **Question (8 pts)**:
   Consider a base station that records the number of users connected to it at discrete intervals. For example, the following sequence of symbols is observed

$$x = [10, 12, 20, 15, 30, 35, 40, 46] \tag{1}$$

   (a) Encode the sequence with PCM, using the following quantizer:

$$\begin{aligned} I_1 &= [0, 10), & y_1 &= 0 \\ I_2 &= [10, 20), & y_2 &= 10 \\ I_3 &= [20, 30), & y_3 &= 20 \\ I_4 &= [30, 40), & y_4 &= 30 \\ I_5 &= [40, +\infty), & y_5 &= 40 \end{aligned} \tag{2}$$

   Compute the MSE and the rate. For the latter, use an approximation that adopts Shannon's coding with the following pmf.

y_k	Pr.
0	0.3
10	0.3
20	0.2
30	0.1
40	0.1

   (b) Encode the sequence with DPCM, using the following quantizer (hint: use PCM for the first sample with the quantizer in (a)):

$$\begin{aligned} I_1 &= (-\infty, -15), & e_1 &= -20 \\ I_2 &= [-15, -5), & e_2 &= -10 \\ I_3 &= [-5, 5), & e_3 &= 0 \\ I_4 &= [5, 15), & e_4 &= +10 \\ I_5 &= [15, +\infty), & e_5 &= +20 \end{aligned} \tag{3}$$

   Compute the MSE and the rate. For the latter, use an approximation that adopts Shannon's coding with the following pmf.

e_k	Pr.
-20	0.01
-10	0.04
0	0.9
+10	0.04
+20	0.01

Solution

(a) Encoding with PCM results in the following signal

$$\tilde{x} = [10, 10, 20, 10, 30, 30, 40, 40] \tag{4}$$

The MSE is equal to 11.25.

According to Shannon's coding the length of each codeword is equal to $\lceil -\log_2 p(\alpha) \rceil$

y_k	l_S
0	2
10	2
20	3
30	4
40	4

The rate is equal to

$$R_S = 2 + 2 + 3 + 2 + 4 + 4 + 4 + 4 = 25 \text{ bits} \tag{5}$$

(b) DPCM encoding proceeds as follows

x	10	12	20	15	30	35	40	46
e	-	2	10	-5	10	5	0	6
$\tilde{e}$	-	0	10	0	10	10	0	10
$\tilde{x}$	10	10	20	20	30	40	40	50

The MSE is equal to 8.75.

According to Shannon's coding the length of each codeword is equal to $\lceil -\log_2 p(\alpha) \rceil$

The rate is equal to

$$R_S = 2 + 1 + 5 + 1 + 5 + 5 + 1 + 5 = 25 \text{ bits} \tag{6}$$

2. **Question (5 pts)**

(a) Illustrate the block diagram of a lossy image coder, briefly describing the role of each module.

(b) Describe how to encode color images.

Solution

See slide sets.

3. **Question (5 pts)**

Discuss the design of the perceptual model in audio coding.

Solution

See slide sets.

y_k	l_S
-20	7
-10	5
0	1
10	5
20	7

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**
   (a) Let X denote a discrete source with M symbols.
   - [T] [F] A uniquely decodable code always uses less than $\log_2 M$ bits / symbols
     FALSE. It is possible to construct uniquely decodable codes that are not efficient, requiring more than $\log_2 M$ bits.
   - [T] [F] If $H_N(X) = H(X)$, $\forall N$, then X is a source without memory.
     TRUE.
   - [T] [F] Any instantaneous code is also uniquely decodable.
     TRUE.
   - [T] [F] $H(X, X) = 2H(X)$.
     FALSE. $H(X, X) = H(X) + H(X|X) = H(X)$

   (b) Consider a speech coder
   - [T] [F] In an open loop vocoder, the encoder embeds the decoder.
     FALSE.
   - [T] [F] The long term predictor accounts for the *pitch structure*.
     TRUE.
   - [T] [F] Let $A_L(z)$ denote the long term predictor. Then, if the filter $1/(1 - A_L(z))$ is fed with white noise, the output is periodic.
     TRUE. The shaping filter $1/(1 - A_L(z))$ reconstructs the *pitch structure* of the signal.

   (c) Consider a DPCM coder.
   - [T] [F] If $\Delta_1 < \Delta_2$, then $D(\Delta_1) < D(\Delta_2)$, where $D(\Delta)$ is the MSE distortion obtained with a uniform quantizer with step size Δ
     TRUE.
   - [T] [F] The coding efficiency of DPCM is always better than, or equal to, that of PCM on any signal.
     FALSE. On average, the coding efficiency is no worse than that of PCM. However, for an individual signal, it might happen that PCM is more efficient than DPCM.
   - [T] [F] The predictor used in DPCM is constrained to be a linear predictor.
     FALSE. Although linear predictors are by far the most popular choice, any predictor can be adopted in DPCM.

5. **Question (Matlab - 10pts)**
   Load the files *f1.mat* and *f2.mat* which contain two consecutive frames of a video sequence. Apply the following to the difference between the two frames.
   (a) Divide in 8×8 blocks and compute the 2D DCT of each block.
   (b) Plot the PSNR-Δ function obtained by quantizing the DCT coefficients, with $\Delta \in \{1, 2, 4, 8, 16\}$

(c) Divide in 8×8 blocks and compute the 2D Hadamard transform of each block. The 2D Hadamard transform is a separable transform that can be obtained from the 1D Hadamard transform (hint: the 1D Hadamard transform matrix is obtained by means of *normc(hadamard(8))*).

(d) Plot the PSNR-Δ function obtained by quantizing the Hadamard coefficients, with $\Delta \in \{1, 2, 4, 8, 16\}$

Solution

```
clear
clc


% Load the files f1.mat and f2.mat which contain consecutive frames of a
% video sequence.
load f1.mat
load f2.mat

% Compute the frame difference
D = f2 - f1;

% Compute the 2D DCT
B = 8; % block size
Delta = [1, 2, 4, 8, 16, 32];
[M, N] = size(D);


for d = 1:length(Delta)
    for bm = 1:M/B
        for bn = 1:N/B

            block = D((bm-1)*B + 1 : (bm-1)*B + B, (bn-1)*B + 1 : (bn-1)*B + B);

            block_DCT = dct2(block);

            block_DCT_q = Delta(d)*round(block_DCT / Delta(d));

            block_q = idct2(block_DCT_q);

            D_q((bm-1)*B + 1 : (bm-1)*B + B, (bn-1)*B + 1 : (bn-1)*B + B) = block_q;

        end
    end

    MSE = (1/(M*N))*sum(sum(((D - D_q).^2)));
    PSNR_DCT(d) = 10*log10(255^2 / MSE);

end

plot(Delta, PSNR_DCT);
xlabel('\Delta');
ylabel('PSNR');

H = hadamard(B);
H = normc(H);


for d = 1:length(Delta)
    for bm = 1:M/B
        for bn = 1:N/B

            block = D((bm-1)*B + 1 : (bm-1)*B + B, (bn-1)*B + 1 : (bn-1)*B + B);

            block_H = H'*block*H;

            block_H_q = Delta(d)*round(block_H / Delta(d));

            block_q = H*block_H_q*H';

            D_q((bm-1)*B + 1 : (bm-1)*B + B, (bn-1)*B + 1 : (bn-1)*B + B) = block_q;

        end
    end

    MSE = (1/(M*N))*sum(sum(((D - D_q).^2)));
    PSNR_H(d) = 10*log10(255^2 / MSE);

end

hold on
plot(Delta, PSNR_H, 'r');
```